Session 1: Docker Fundamentals

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Course Materials

https://github.com/mcpp-volodymyr/docker-workshop

- Slides, examples, and commands from this session are available in the repository
- No detailed note-taking required
- Focus on understanding the concepts and asking questions
- Start with README.md

Agenda

- The problem Docker solves
- Docker architecture overview
- Images, containers, and registries
- Essential Docker commands
- Building and running a container
- Practice task
- Questions

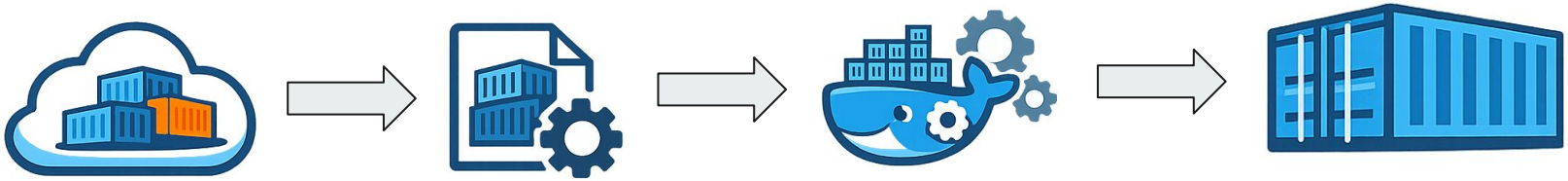
What Is Docker?

Running the same software on different machines is like cooking the same recipe in different kitchens.

- Software depends on more than just source code
- Different environments can produce different results
- Docker packages an application together with what it needs to run
- This makes environments easier to reproduce and share

Docker Architecture Overview

- A **registry** stores images
- The **Docker engine** pulls **images** from a registry
- The engine creates and runs **containers** from those images
- A container is a running instance of an image



Docker on Different Operating Systems

Platform	Linux Containers	Windows Containers	macOS Containers
Linux	✅ Yes	❌ No	❌ No
Windows	✅ Yes	✅ Yes	❌ No
macOS	✅ Yes	❌ No	❌ No

- macOS containers are not supported
- Containers are OS-specific, and images also target a specific CPU architecture such as amd64 or arm64

What Is an Image?

- A Docker image is a **read-only template**
- It contains the application and what it needs to run
- Images are used to create containers
- Multiple containers can be created from the same image

What Is a Container?

- A container is a **running instance of an image**
- It runs in an isolated environment
- Containers can be started, stopped, and removed
- Containers are created from images

What Is a Registry?

- A registry stores and distributes Docker images
- Docker can pull images from a registry
- You can also push your own images to a registry
- Docker Hub is a common example

Working with Images

Manage images on your machine

- *docker images*
- *docker pull*
- *docker build*
- *docker rmi*

Working with Containers

Run and manage containers

- *docker run*
- *docker ps*
- *docker stop*
- *docker start*
- *docker rm*

Working with Registries

Authenticate and transfer images

- *docker login*
- *docker pull*
- *docker push*
- *docker logout*

Running Hello World

A good first step after a clean Docker installation is to pull and run the **hello-world** image.

docker images

docker pull hello-world

docker images

docker run hello-world

docker ps -a

What Is a Dockerfile?

- A **Dockerfile** is a text file with instructions for building a Docker image
- *docker build* uses a Dockerfile to create an image

Dockerfile Example

```
FROM python:3.12-slim

ENV PYTHONUNBUFFERED=1

WORKDIR /app
COPY server.py .
EXPOSE 8000

CMD ["python", "server.py"]
```

FROM

FROM python:3.12-slim

- base image
- starting point of our image

ENV

ENV PYTHONUNBUFFERED=1

- sets environment variables inside the image
- here it makes Python logs appear immediately

WORKDIR

WORKDIR /app

- sets working directory inside the image
- following commands use this location

COPY

COPY server.py .

- copies files from the build context into the image
- destination relative to the current WORKDIR

EXPOSE

EXPOSE 8000

- documents that the application uses port 8000
- does not publish the port to the host

CMD

CMD ["python", "server.py"]

- default command to run when the container starts

Build and Run the Image

docker build -t demo-http .

docker run --rm -p 8000:8000 demo-http

Useful Links

- https://github.com/mcpp-volodymyr/docker-workshop

  Session materials, slides, examples, and practice tasks

- https://docs.docker.com/reference/dockerfile/

  Dockerfile reference

- https://docs.docker.com/reference/cli/docker/

  Docker CLI reference

- https://hub.docker.com/

  Public Docker image catalog

Practice Task

- Follow the instructions in session-1/homework.md
- Use the repository materials as reference
- Bring your questions to the next session

Thank You!